

AMAL THILAKAN

☎ +91 8921654839 | @ amalthilakan111@gmail.com |  LinkedIn |  Portfolio |  GitHub

SUMMARY

Master's graduate in Computer Science with a specialization in Data Science, experienced in full-stack web and mobile application development, machine learning, and data-driven system design. Proficient in frontend and backend technologies, RESTful APIs, database design, and scalable application architecture. Hands-on experience working in Agile/Scrum environments, collaborating with cross-functional teams, participating in sprint planning, code reviews, and iterative development. Strong problem-solving and communication skills with a solid understanding of the software development lifecycle (SDLC), clean code practices, performance optimization, and project management.

EXPERIENCE

AY Tech

Python Django Intern

Kerala, India

Sept 2025 – Present

- Developed scalable backend services using Django and Django REST Framework following RESTful API standards.
- Designed and optimized database schemas using MySQL and ORM-based query optimization techniques.
- Implemented authentication, authorization, and role-based access control systems.
- Collaborated in Agile/Scrum teams, participating in sprint planning, daily stand-ups, and code reviews.
- Integrated machine learning workflows and AI-driven features into backend services.

Centgrade Labs

Software Engineer Intern

Kerala, India

Oct 2024 – Feb 2025

- Contributed to full-stack development using React, Node.js, Express.js, and MongoDB.
- Developed reusable UI components and integrated REST APIs for frontend-backend communication.
- Worked closely with senior engineers to build scalable, maintainable application modules.
- Followed Agile development practices and version control workflows using Git.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++

Frontend Technologies: HTML, CSS, React, React Native, Redux, Axios

Backend Technologies: Django, Django REST Framework, Node.js, Express.js

Databases: MySQL, SQLite, MongoDB, SQL

Machine Learning & Data Science: Pandas, NumPy, Scikit-learn, Data Analysis, Model Building

Tools & Platforms: Git, GitHub, VS Code, Expo

Methodologies & Practices: Agile, Scrum, SDLC, RESTful Design, MVC Architecture, Clean Code

PROJECTS

VHELP Platform

Tools & Technologies: Python, Django, SQLite, HTML, CSS, JavaScript, AI, Machine Learning

- Designed and developed a role-based volunteer-consumer assistance platform with secure authentication.
- Implemented multi-step registration workflows, admin dashboards, and approval systems.
- Applied machine learning techniques to improve volunteer-to-case matching accuracy.

FoodPin App

Tools & Technologies: React Native, Expo, Google Maps API, TypeScript

- Developed an Android app to bookmark restaurants with map integration.
- Implemented geolocation, routing, and list management features.

Recipe Recommendation System

Tools & Technologies: MERN Stack, Machine Learning

- Built a personalized recommendation system using machine learning algorithms and user preferences.

Indian Sign Language Detection

Tools & Technologies: Python, MediaPipe, OpenCV, RNN

- Developed a real-time Indian sign language recognition system using computer vision and neural networks.

Breast Cancer Detection System

Tools & Technologies: Python, Pandas, NumPy, Scikit-learn

- Implemented predictive machine learning models for early-stage breast cancer classification.

EDUCATION

Master of Computer Science (Data Science)

Cochin University of Science and Technology

2022 – 2024

Kallassery, Kerala, India

Bachelor of Computer Applications (BCA)

MES Marampally, Mahatma Gandhi University

2018 – 2021

Marampally, Kerala, India

Higher Secondary Education (Science)

Kasturba English Medium School (CBSE)

2016 – 2018

Manjummel, Kerala, India

SSLC

Kasturba English Medium School (CBSE)

2014 – 2016

Manjummel, Kerala, India

CERTIFICATES

MEARN Stack Development

Right Soft Options, Kaloor

Jun 2024 – Sep 2024

Android Development Using Java

Acmegrade, Bengaluru

Jun 2023 – Jul 2023